

EXPERIMENTS

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COURSE CODE: **DSA0602 DATA HANDLING AND VISUALIZATION**

Set 7 – Customer Demographics Analysis

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

QUESTION 1

Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

AIM

To create a bar chart to visualize the distribution of customer ages using R.

PROCEDURE

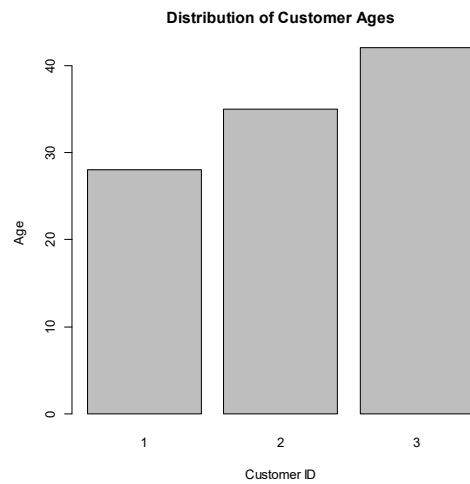
1. Create the customer demographics dataset.
2. Store the Customer ID and Age values.
3. Use the barplot() function to create the bar chart.
4. Label the X-axis and Y-axis.
5. Add a suitable title.
6. Display the chart.

CODE

```
customer <- data.frame(  
  Customer_ID = c(1, 2, 3),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),  
  Income = c(50000, 60000, 75000)  
)  
  
barplot(  
  customer$Age,
```

```
names.arg = customer$Customer_ID,  
xlab = "Customer ID",  
ylab = "Age",  
main = "Distribution of Customer Ages"  
)
```

OUTPUT



RESULT

Thus, the bar chart representing the distribution of customer ages was successfully created.

QUESTION 2

Generate a pie chart to represent the distribution of customers by gender.

AIM

To create a pie chart representing the distribution of customers based on gender.

PROCEDURE

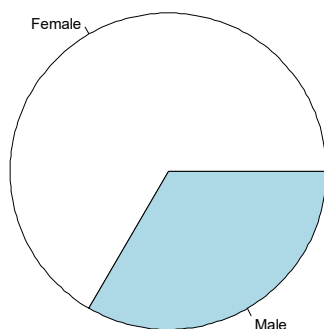
1. Create the customer demographics dataset.
2. Count the number of customers in each gender.
3. Use the table() function.
4. Use the pie() function to create the pie chart.
5. Add a suitable title.
6. Display the chart.

CODE

```
customer <- data.frame(  
  Customer_ID = c(1, 2, 3),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),  
  Income = c(50000, 60000, 75000)  
)  
  
gender_count <- table(customer$Gender)  
  
pie(  
  gender_count,  
  labels = names(gender_count),  
  main = "Customer Distribution by Gender"  
)
```

OUTPUT

Customer Distribution by Gender



RESULT

Thus, the stacked bar chart was successfully created to represent product quantities within different categories.

QUESTION 3

Build a table to show the demographic information for each customer. Label the table elements.

AIM

To create a table displaying the demographic information of each customer.

PROCEDURE

1. Create the customer demographics dataset.
2. Enter Customer ID, Age, Gender and Income.
3. Assign suitable column names.
4. Display the dataset as a table.

CODE

```
customer <- data.frame(  
  Customer_ID = c(1, 2, 3),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),  
  Income = c(50000, 60000, 75000)  
)
```

```
print(customer)
```

OUTPUT

```
Customer_ID Age Gender Income  
1          1  28 Female  50000  
2          2  35   Male  60000  
3          3  42 Female  75000
```

RESULT

Thus, the scatter plot was successfully created to analyze the relationship between product price and quantity available.

QUESTION 4

Develop an interactive dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

AIM

To develop an interactive customer demographics dashboard using R Shiny by combining a bar chart, pie chart and customer information table.

PROCEDURE

1. Create the customer demographics dataset.
2. Load the Shiny package.
3. Create a user interface containing the bar chart, pie chart and table.
4. Create the server section to generate the charts and table.
5. Add a gender filter for interactive exploration.
6. Run the Shiny application.
7. View the dashboard in the browser.

CODE

```
library(shiny)

customer <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

ui <- fluidPage(
  titlePanel("Customer Demographics Dashboard"),

  sidebarLayout(
    sidebarPanel(
      selectInput(
        "gender",
        "Select Gender:",
        choices = c("All", "Female", "Male")
      )
    ),

    mainPanel(
      plotOutput("agePlot"),
      plotOutput("genderPlot"),
      tableOutput("customerTable")
    )
  )
)
```

```

server <- function(input, output) {

  filteredData <- reactive({
    if (input$gender == "All") {
      customer
    } else {
      customer[customer$Gender == input$gender, ]
    }
  })

  output$agePlot <- renderPlot({
    data <- filteredData()

    barplot(
      data$Age,
      names.arg = data$Customer_ID,
      xlab = "Customer ID",
      ylab = "Age",
      main = "Distribution of Customer Ages"
    )
  })

  output$genderPlot <- renderPlot({
    data <- filteredData()

    gender_count <- table(data$Gender)

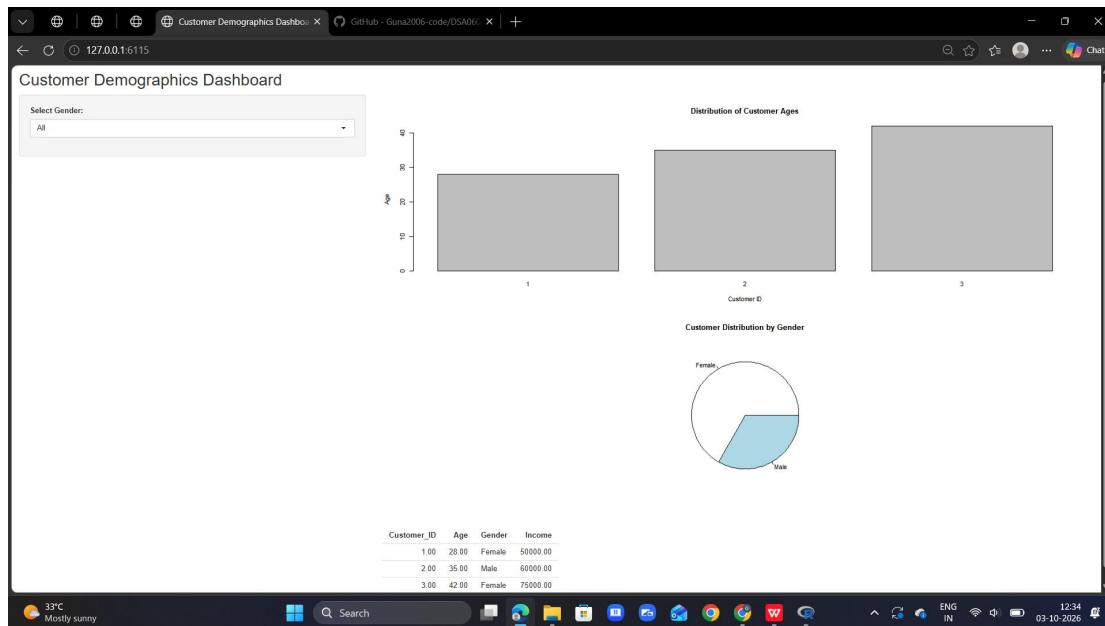
    pie(
      gender_count,
      labels = names(gender_count),
      main = "Customer Distribution by Gender"
    )
  })

  output$customerTable <- renderTable({
    filteredData()
  })
}

shinyApp(ui = ui, server = server)

```

OUTPUT



RESULT

Thus, an interactive customer demographics dashboard containing a bar chart, pie chart and demographic table was successfully created using **R Shiny**.